

a) BRAND OR PRODUCT NAME

Insulinagyp N 100 IU/mL (NPH Human Insulin Isophane Suspension) in 3mL Cartridge

b) NAME AND STRENGTH OF ACTIVE SUBSTANCE

b.1 Active Substance

Insulin Isophane NPH 100 IU/mL (rDNA Origin).

c) PRODUCT DESCRIPTION

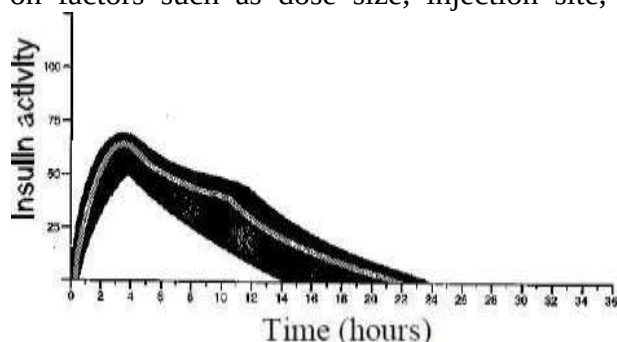
Insulinagyp N, is a white suspension which, on standing, deposits a white sediment and leaves a colorless or almost colorless supernatant liquid. The sediment is readily resuspended by gentle shaking. The product is filled into Type I glass cartridges containing a glass bead and sealed with rubber closures consisting of a rubber plunger at the bottom and a laminated rubber disc seal at the top of the cartridge.

d) PHARMACODYNAMICS

The primary activity of insulin is the regulation of glucose metabolism.

In addition, insulin has several anabolic and anti-catabolic actions on a variety of different tissues. Within muscle tissue, this includes increasing glycogen, fatty acid, glycerol, and protein synthesis and amino acid uptake, while decreasing glycogenolysis, gluconeogenesis, ketogenesis, lipolysis, protein catabolism, and amino acid output.

The typical activity profile (glucose utilization curve) following subcutaneous injection is illustrated below by the heavy line. Variations that a patient may experience in timing and/or intensity of insulin activity are illustrated by the shaded area. Individual variability will depend on factors such as dose size, injection site, temperature, and physical activity of the patient.



e) PHARMACOKINETIC

The pharmacokinetics of insulin do not reflect the metabolic action of that hormone. Therefore it is more appropriate to examine glucose utilisation curves (as discussed above) when considering the activity of insulin.

f) INDICATION

For the treatment of patients with diabetes mellitus for whom diet and/or oral agents are not sufficient.

g) RECOMMENDED DOSAGE

Posology

The dosage should be determined by the physician, according to the requirements of the patient.

Paediatric population

No data are available

Method of administration

Subcutaneous administration should be in the upper arms, thighs, buttocks, or abdomen. Use of injection sites should be rotated so that the same site is not used more than approximately once a month in order to reduce the risk of lipodystrophy and cutaneous amyloidosis.

Care should be taken when injecting any Insulinagyp preparation to ensure that a blood vessel has not been entered. After any insulin injection, the injection site should not be massaged. Patients must be educated to use proper injection techniques.

Insulinagyp N in cartridges is only suitable for subcutaneous injections from a reusable pen. This formulation should not be administered intravenously.

h) ROUTE OF ADMINISTRATION

Subcutaneous

i) CONTRAINDICATIONS

Insulinagyp N is contraindicated in hypoglycemia. Human insulin is contraindicated in patients with hypersensitivity to human insulin or any of its excipients (unless used as part of a desensitization program).

j) WARNINGS AND PRECAUTIONS

Transferring a patient to another type or brand of insulin should be done under strict medical supervision. Changes in strength, brand (manufacturer), type (soluble, isophane, mixture), species (animal, human, human insulin analog), and/or method of manufacture (recombinant DNA versus animal-source insulin) may result in the need for a change in dosage.

Some patients taking human insulin may require a change in dosage from that used with animal-source insulins. If an adjustment is needed, it may occur with the first dose or during the first several weeks or months.

A few patients who experienced hypoglycemic reactions after transfer to human insulin have reported that the early warning symptoms were less pronounced or different from those experienced with their previous animal insulin. Patients whose blood glucose is greatly improved, e.g., by intensified insulin therapy, may lose some or all of the warning symptoms of hypoglycemia and should be advised accordingly. Other conditions that may make the early warning symptoms of hypoglycemia different or less pronounced include long duration of diabetes, diabetic nerve disease, or medications such as beta blockers. Uncorrected hypoglycemic and hyperglycemic reactions can cause loss of consciousness, coma, or death.

The use of dosages that are inadequate or discontinuation of treatment, especially in insulin-dependent diabetics, may lead to hyperglycemia and diabetic ketoacidosis, which are potentially lethal.

Treatment with human insulin may cause the formation of antibodies, but antibody titers are lower than those to purified animal insulin.

Insulin requirements may change significantly in diseases of the adrenal, pituitary, or thyroid glands and in the presence of renal or hepatic impairment.

Insulin requirements may be increased during illness or emotional disturbances.

Adjustment of insulin dosage may also be necessary if patients change their level of physical activity or their usual diet.

patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycemia. Blood glucose monitoring is recommended after the change in the injection site, and dose adjustment of antidiabetic medications may be considered.

Combination of Human Insulin with Thiazolidinedione

Cases of cardiac failure have been reported when thiazolidinedione was used in combination with insulin, especially in patients with risk factors for development of congestive heart failure. This should be kept in mind if treatment with the combination of thiazolidinedione and human insulin is considered. If the combination is used, patients should be observed for signs and symptoms of heart failure, weight gain, and edema. Thiazolidinedione should be discontinued if any deterioration in cardiac symptoms occurs.

Skin and subcutaneous tissue disorders

Patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycaemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycaemia. Blood glucose monitoring is recommended after the change in the injection site and dose adjustment of antidiabetic medications may be considered.

k) INTERACTIONS WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

A number of medicinal products are known to interact with glucose metabolism and therefore the physician should be consulted when using other medications in addition to human insulin. The physician must take possible interactions into account and should always ask patients about any medicinal products they take.

Insulin requirements may be increased by substances with hyperglycemic activity, such as glucocorticoids, thyroid hormones, growth hormone, danazol, beta2-sympathomimetics (such as ritodrine, salbutamol, terbutaline), and thiazides.

Insulin requirements may be reduced in the presence of substances with hypoglycemic activity, such as oral hypoglycemics (OHA), salicylates (e.g., acetylsalicylic acid), certain antidepressants

(monoamine oxidase inhibitors), certain angiotensin-converting enzyme (ACE) inhibitors (e.g., captopril, enalapril), angiotensin II receptor blockers, non-selective beta- blocking agents, and alcohol. Somatostatin analogs (octreotide, lanreotide) may either decrease or increase insulin dose requirements.

l) PREGNANCY AND LACTATION

Pregnancy

It is essential to maintain good control of the insulin-treated (insulin-dependent or gestational diabetes) patient throughout pregnancy. Insulin requirement usually decreases during the first trimester of pregnancy and increases during the second and third trimesters. Women with diabetes should be advised to inform their doctors if they are pregnant or are contemplating pregnancy.

Careful monitoring of glucose control, as well as general health, is essential in pregnant patients with diabetes.

Breast-feeding

Women with diabetes who are lactating may require adjustments in insulin dose and/or diet.

m) SIDE EFFECTS

Hypoglycaemia is the most frequent undesirable effect of insulin therapy that a patient with diabetes may suffer. Severe hypoglycaemia may lead to loss of consciousness, and in extreme cases, death. No specific frequency for hypoglycaemia is presented, since hypoglycaemia is a result of both the insulin dose and other factors e.g., a patient-related factors such as diet and exercise levels

Local allergy in patients is common ($\geq 1/100$ to $< 1/10$).

Redness, swelling, and itching can occur at the site of insulin injection. This condition usually resolves in a few days to a few weeks. In some instances, local reactions may be related to factors other than insulin, such as irritants in the skin cleansing agent or poor injection technique.

Systemic allergy, which is very rare ($< 1/10,000$) but potentially more serious, is generalized allergy to insulin. It may cause rash over the whole body, shortness of breath, wheezing, reduction in blood pressure, fast pulse, or sweating. Severe cases of generalized allergy may be life-threatening. In the rare event of a severe allergy to Insulinaglypt N, treatment is required immediately. A change of insulin or desensitization may be required.

Lipodystrophy at the injection site is uncommon ($\geq 1/1,000$ to $< 1/100$).

Skin and subcutaneous tissue disorders: Frequency “unknown”: Cutaneous amyloidosis.

Skin and subcutaneous tissue disorders: Lipodystrophy and cutaneous amyloidosis may occur at the injection site and delay local insulin absorption. Continuous rotation of the injection site within the given injection area may help to reduce or prevent these reactions.

Cases of oedema have been reported with insulin therapy, particularly if previous poor metabolic control is improved by intensified insulin therapy.

n) SYMPTOMS AND TREATMENT OF OVERDOSE

Insulin has no specific overdose definition because serum glucose concentrations are the result of complex interactions between insulin levels, glucose availability, and other metabolic processes.

Hypoglycemia may occur as a result of an excess of insulin relative to food intake and energy expenditure.

It may be associated with listlessness, confusion, palpitations, headache, sweating, and vomiting.

Mild hypoglycemic episodes will respond to oral administration of glucose or sugar products.

Moderately severe hypoglycemia can be corrected by intramuscular or subcutaneous administration of glucagon, followed by oral carbohydrate when the patient recovers sufficiently.

Patients who fail to respond to glucagon must be given intravenous glucose solution.

If the patient is comatose, glucagon should be administered intramuscularly or subcutaneously. However, intravenous glucose must be given if glucagon is not available or the patient fails to respond to glucagon. The patient should be given a meal as soon as consciousness is recovered.

Sustained carbohydrate intake and observation may be necessary because hypoglycemia may occur after apparent clinical recovery.

o) EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Your ability to concentrate and react may be reduced if you experience hypoglycaemia (low blood sugar).

Please remember this in all situations where you may endanger yourself and others (e.g., driving a car or operating machinery). You should contact your diabetes nurse or doctor about driving if you have:

- frequent episodes of hypoglycemia
- reduction or absence of hypoglycemia warning signs.

p) PRECLINICAL SAFETY DATA

Insulinaglypt N is human insulin produced by recombinant technology. No serious events have been reported in sub chronic toxicology studies. Human insulin was not mutagenic in a series of in vitro and in vivo genetic toxicity assays.

q) INSTRUCTION FOR USE

Do not reuse needles. Dispose of the needle in a responsible manner. Needles must not be shared.

Cartridges can be used until empty, then properly discard. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Instructions for use and handling:

A solution for injection in a 3mL cartridge to be used in conjunction with reusable pen only. Please refer to the User Manual of the reusable pen for more information.

a. Preparing a dose

Before using the reusable pen, the user manual must be read carefully. The reusable pen must be used as recommended in the user manual. Pens should not be used if any part looks broken or damaged.

The cartridges should be examined frequently and should not be used if clumps of material are present or if solid white particles stick to the bottom or wall of the cartridge, giving a frosted appearance.

b. Injecting a dose

Refer to the detailed instructions for preparing the pen and injecting the dose, the following is a general description.

1. Wash your hands.
2. Choose a site for injection.
3. Clean the skin as instructed.
4. Stabilize the skin by spreading it or pinching up a large area. Insert the needle and inject as instructed.
5. Pull the needle out and apply gentle pressure over the injection site for several seconds. Do not rub the area.
6. For an injection device use the outer needle cap, unscrew the needle and dispose of it safely.
7. Use of the injection sites should be rotated so that the same is not used more than approximately once a month.

r) SPECIAL PRECAUTIONS FOR DISPOSAL AND OTHER HANDLING

Do not reuse needles. Dispose of the needle in a responsible manner. Needles must not be shared. Cartridge can be used until empty, then properly discard. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

s) STORAGE CONDITIONS

Not in use (unopened): should be stored in a refrigerator at temperature (2-8 °C), do not freeze. In-use (opened): can be kept unrefrigerated as long as it is kept as cool as possible, below (30°C), away from heat and light.

t) DOSAGE FORM AND PACKAGING AVAILABLE

Product is filled into Type I glass cartridges, containing a glass bead and sealed with rubber closures which consists of rubber plunger at the bottom and laminated rubber disc seal at the top of the cartridge.

Available in a box of 5 x 3 mL cartridges.

u) PHARMACEUTICAL PARTICULARS

u.1 Excipients

Meta cresol Distilled	1.6 mg
Glycerol	16 mg
Liquefied Phenol	0.73 mg
(equivalent to Phenol)	0.65 mg
Protamine Sulfate	0.244 mg
Dibasic Sodium Phosphate	3.78 mg
Zinc Oxide	Approx. 9 µg
Hydrochloric Acid 10%	q.s.
Sodium Hydroxide 10%	q.s.
Water for injection	q.s. to make 1 mL

u.2 Shelf Life

Unopened cartridges 24 months

After first use 28 days

u.3 Incompatibilities

Insulinagyp N must not be mixed with insulins produced by other manufacturers or with animal insulin preparation.

v) NAME AND ADDRESS OF MANUFACTURER/PRODUCT REGISTRATION HOLDER**v.1 MANUFACTURED AND RELEASED BY**

Medical Union Pharmaceuticals (MUP), Insulin plant, part no (1) – 2nd industrial area-6th October City-Giza-Egypt

v.2 PRODUCT REGISTRATION HOLDER

DMW International SDN. BHD.
NO, B2-09, 2ND Floor (Lift 5), Megan Embassy,
Block B 225 Jalan Ampang
Kuala Lumpur, Malaysia

w) DATE OF REVISION

11/05/2025